

EXAMPLE TEMPLATE
GUIDE

BY

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THESIS

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Abstract

Abstract.

To whomever you want.

Acknowledgments

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τ	Time taken to drink one cup of coffee.
μg	Micrograms (of caffeine, generally).

Chapter 1

Introduction

Nuclear power has supplied the United States with large quantities of low-carbon electricity for more than six decades. The fuel discharged from those reactors, however, remains a long-term national responsibility. By the end of 2022, commercial reactors in the United States had permanently discharged roughly 91,000 metric tons of uranium (MTU) of used nuclear fuel (UNF), and the inventory continues to grow by approximately 2,000 MTU each year [1]. This material is distributed across more than one hundred sites in thirty-nine states, including operating nuclear power plants, decommissioned reactor sites, and independent spent fuel storage installations. Because the United States does not yet operate a permanent geologic repository for commercial spent fuel, most of the inventory remains stored at or near the sites where it was generated [2].

The challenge is not simply that the inventory is large. It is also highly heterogeneous. The phrase “the U.S. used nuclear fuel inventory” can suggest a single stockpile with one representative composition, but the actual inventory is the accumulated result of decades of reactor operation. Assemblies discharged in different years were fabricated with different initial enrichments, irradiated to different burnups, operated in different reactor types, and cooled for different lengths of time. Their remaining heavy-metal masses and isotopic compositions therefore vary considerably. Some assemblies contain larger quantities of residual fissile material, plutonium, or minor actinides than others. Some have cooled for several decades, while others were discharged recently and continue to generate substantial decay heat. Any strategy that attempts to process, recycle, or transmute this material must eventually confront those differences.

UNF also changes with time. During the first decades after discharge, its heat generation and radiation field are strongly influenced by fission products. As those nuclides decay, the relative importance of plutonium and the minor actinides increases. Their long half-lives contribute to the need to isolate untreated fuel over timescales on the order of 10^5 years [3]. Among the long-lived components of used fuel, the actinides are distinctive because they can be destroyed through fission rather than managed only through continued storage and radioactive decay. That possibility does not eliminate the need for a repository, nor does it remove every radioactive waste stream. It does, however, create the possibility of changing the quantity and composition of the material that must ultimately be isolated.

Accelerator-driven system (ADS) have been proposed as one technology for carrying out this transmutation. An ADS couples an externally generated neutron source to a subcritical multiplying core. Because the core remains below criticality, its operation depends on the external source, allowing fuel compositions with high concentrations of minor actinides to be considered under conditions that would be more difficult to accommodate in a critical reactor. Detailed studies have examined accelerator reliability, spallation targets,

subcritical-core physics, liquid-metal coolants, fuel behavior, and the rates at which plutonium and minor actinides could be consumed. These studies establish the physical basis for accelerator-driven transmutation and provide increasingly realistic designs for future facilities.

A technically feasible core, however, is not the same as a viable transmutation strategy. A reactor-physics calculation may show how much actinide material one system can consume, but it does not determine where that material will come from, which assemblies will be processed first, how the feed will change over time, or how many facilities will be required. Before an ADS can consume actinides, spent fuel must be selected from storage, transported, cooled as needed, separated, and converted into a suitable feed form. The transmutation facility must then be deployed, supplied, operated, and eventually integrated with storage, recycle, and waste-management facilities. The material discharged from the ADS may require further processing or repeated recycle before the desired degree of actinide destruction is achieved.

The central problem is therefore a fuel-cycle problem rather than only a single-core problem. It requires a model that can follow material from the existing inventory, through selection and processing, into a transmutation facility, and finally into the inventories and waste streams that remain. Such a model must operate over timescales of years or decades while retaining enough information to distinguish among assemblies. It must also allow facilities to compete for resources, enforce throughput limits, and respond to changes in material availability. These capabilities are difficult to represent using a single average spent-fuel composition.

Many previous fuel-cycle studies have necessarily simplified the inventory by representing it as an aggregated material stream or a small set of representative recipes. That approach is useful when comparing generic future fuel cycles or studying equilibrium reactor fleets. It is less satisfactory when the starting point is the existing U.S. inventory. A national-average recipe cannot identify whether the material it represents comes from older or newer assemblies, whether it has high or low burnup, whether it is concentrated at a few sites or dispersed across the country, or which assemblies would actually be selected under a particular management strategy. These differences may affect the amount and quality of material available for transmutation and the characteristics of the inventory that remains untreated.

Recent inventory databases make a more realistic representation possible. The STANDARDS 5.0.1 database contains assembly-level information describing the U.S. commercial spent-fuel inventory, including attributes such as discharge date, burnup, initial enrichment, mass, and calculated isotopic composition. This level of detail creates an opportunity to treat the national inventory not as a single homogeneous stockpile, but as a collection of discrete resources that can be selected according to explicit criteria. The database alone, however, does not describe how those assemblies would move through a future fuel cycle. The information must be incorporated into a dynamic simulator capable of representing transactions among facilities.

The work is conducted within the Advanced Reactors and Fuel Cycles group at University of Illinois Urbana-Champaign (UIUC), where agent-based fuel-cycle simulation is used to examine the system-level consequences of advanced nuclear technologies. It also contributes to the UIUC-led Enhanced Integrated Nuclear Systems for Transmutation and Efficient Isolation of Nuclides (EINSTEIN) project under the Advanced Research Projects Agency-Energy (ARPA-E) Nuclear Energy Waste Transmutation Optimized Now (NEWTON) program, which brings together neutronics, nuclear-data, materials, fuel-cycle, and techno-economic analyses to evaluate externally driven transmutation systems. My own research in the group has involved working directly with CYCLUS, detailed spent-fuel inventory data, and higher-fidelity reactor calculations, which made the disconnect between the available inventory information and existing system-level transmutation models especially clear. That combination of experience placed me in a position to build the missing connection between the fuel assemblies that exist today and the transmutation technologies being

proposed for the future.

This thesis uses the CYCLUS nuclear fuel cycle simulation framework to make that connection. CYCLUS represents the fuel cycle as a system of interacting agents, including regions, institutions, and facilities. Individual facilities request, offer, and exchange material through the Dynamic Resource Exchange, while modular software components known as archetypes define their behavior. This structure makes it possible to represent the U.S. inventory as an active supplier rather than as a static input file. It also allows that supplier to interact with processing, storage, and transmutation facilities within the same simulation.

Two new facility archetypes are developed for this purpose. The `us_inventory` archetype represents individual U.S. spent-fuel assemblies as selectable material resources. Each assembly retains metadata including its heavy-metal mass, discharge date, burnup, initial enrichment, and isotopic composition derived from Storage Transportation and Disposal Analysis Resource and Data System. (STANDARDS) 5.0.1 [4], [5]. Rather than collapsing these records into one average composition, the facility supplies material through the CYCLUS Dynamic Resource Exchange according to user-defined throughput limits and inventory-selection policies.

The second archetype represents an ADS. Its material transformation is based on precomputed Serpent depletion calculations, allowing the fuel-cycle simulation to incorporate higher-fidelity neutronics results without performing a full transport and depletion calculation during every simulation time step. Coupling the two archetypes makes it possible to follow material from individual inventory records into the modeled transmutation system and to observe how the remaining inventory changes as material is withdrawn.

The selection policies considered in this work rank available assemblies according to characteristics such as discharge date, burnup, and initial enrichment. These policies are not intended to represent final operational decisions or recommendations for U.S. waste management. Instead, they provide controlled cases for examining how different assumptions about access to the inventory affect the material supplied to a transmutation system. For example, prioritizing older assemblies may produce a different feed composition than prioritizing high-burnup or high-enrichment assemblies, even when the same total mass is supplied. The simulation records these differences and preserves the connection between the supplied material and the assemblies from which it originated.

The primary contribution of this thesis is a material-flow model rather than a new accelerator-driven reactor design or a full evaluation of whether the United States should deploy accelerator-driven transmutation. A decision of that kind would still require detailed neutronics, thermal-hydraulics, materials, safety, economic, transportation, safeguards, and repository analyses. What this work provides instead is a model that connects a realistic, assembly-resolved representation of the U.S. inventory to an ADS facility inside a dynamic fuel-cycle simulation.

This connection provides a foundation for broader policy and deployment studies. A future analysis could add site-specific transportation, separations capacity, construction schedules, facility siting, operating costs, or alternative transmutation technologies. Those questions cannot be fully grounded in the present inventory, however, unless the model first represents the assemblies that actually exist and the rules by which they become available. By preserving assembly-level differences, this work makes the assumptions behind a transmutation scenario more transparent and its results more traceable.

The broader motivation is to ground the discussion of transmutation in how a real national inventory could supply such facilities over time, rather than in what a single idealized facility can destroy on its own. This emphasis matters because the performance of a technology in isolation may differ from its performance as part of an operating fuel cycle. Through the integration of assembly-level inventory data and an ADS

representation in CYCLUS, this thesis provides one step toward evaluating that larger system.

The remainder of this thesis is organized as follows. Chapter 2 reviews UNF management in the United States, the CYCLUS framework, accelerator-driven systems, the EINSTEIN program, and prior transmutation fuel-cycle simulations. It concludes by identifying the research gap addressed in this work. Chapter 3 describes the design and implementation of the `us_inventory` and ADS archetypes, the preparation of their input data, and the simulation scenarios considered. Chapter 4 presents the simulation results and compares the effects of the alternative inventory-selection policies. Finally, Chapter 5 discusses the implications and limitations of the results and identifies directions for future development.

Chapter 2

Background and Literature Review

2.1 Used Nuclear Fuel in the United States

UNF, also commonly called spent nuclear fuel, is nuclear fuel that has been removed from a reactor because it can no longer efficiently support the desired operating cycle. Although the fuel is no longer used in the reactor, it remains thermally hot and highly radioactive because it contains radioactive fission products, plutonium, and minor actinides produced during irradiation [6]. The term “used” is particularly appropriate because the discharged material is not completely depleted: most of its uranium remains, and the fuel also contains newly generated fissile isotopes that could potentially be recovered and reused.

The radioactivity and heat generation of UNF decrease with time, but they do so over multiple timescales. Short-lived fission products dominate the radiation field and decay heat immediately following discharge. At longer times, plutonium and the minor actinides become increasingly important because many of these isotopes have half-lives ranging from hundreds to tens of thousands of years. Consequently, untreated commercial UNF may require isolation over timescales on the order of 10^5 years before its radiotoxicity approaches that of the natural uranium ore from which the fuel was originally produced [3]. The long duration of this hazard is a central challenge in nuclear waste management.

The internationally accepted approach for the permanent isolation of UNF and high-level radioactive waste is disposal in a deep geologic repository [7]. The United States, however, does not currently operate a permanent repository for commercial UNF. Until such a repository or another disposition pathway becomes available, reactor licensees must store the fuel at or near the sites where it was generated [2], [6]. Newly discharged fuel is first placed in water-filled spent-fuel pools, which provide both cooling and radiation shielding. After several years of cooling, assemblies may be transferred to sealed dry-storage systems at independent spent fuel storage installations.

Because a national disposal system has not yet been implemented[2], the geographic distribution of UNF largely reflects the historical deployment of the U.S. reactor fleet. Most commercial fuel remains at operating or decommissioned nuclear power plant sites, including locations at which the reactor itself has been permanently shut down. The result is a distributed inventory that must eventually be transported, consolidated, processed, or disposed of[1].

As of the end of 2022, approximately 91,000 MTU from commercial pressurized-water reactors and boiling-water reactors had been permanently discharged in the United States. This inventory grows by roughly 2,000 MTU per year as operating reactors discharge fuel [8], and the current reactor fleet is projected

to generate on the order of 140,000 MTU in total by 2060. The U.S. Department of Energy (DOE) managed an additional inventory of approximately 2,446 metric ton of heavy metal (MTHM), excluding most naval fuel. Unlike the comparatively standardized commercial light-water-reactor inventory, DOE-managed UNF includes fuel from defense-production reactors, research and test reactors, early demonstration reactors, and other specialized programs. Altogether, UNF and primary reprocessing wastes were distributed across more than 100 locations in 39 states [1].

Table 2.1 summarizes the scale and distribution of this inventory. Commercial fuel dominates the national inventory by mass, while DOE-managed material is considerably more diverse in enrichment, fuel form, cladding, geometry, and irradiation history.

Table 2.1: The U.S. inventory of used nuclear fuel and primary reprocessing waste was distributed among commercial, DOE, research, and legacy waste sites. Data from [1], [8].

Inventory category	Reported quantity
Commercial UNF at nuclear power plant and ISFSI sites	90,963 MTHM
Permanently discharged PWR fuel	135,645 assemblies; 59,013 MTHM
Permanently discharged BWR fuel	179,466 assemblies; 32,023 MTHM
Commercial dry-storage canisters or casks	3,862
DOE-managed UNF	approximately 2,446 MTHM ^a
UNF at university, government, and commercial research sites	approximately 1 MTHM
Commercial nuclear power plant and ISFSI sites storing UNF	73 sites
Sites with all nuclear power reactors permanently shut down	20 sites
Operating commercial nuclear power reactors	94 reactors
Permanently shut-down commercial nuclear power reactors	27 reactors
DOE sites with UNF storage or research reactors	6 sites
University research-reactor sites	20 sites
Other government research-reactor sites	4 sites
Commercial research and development sites with UNF	4 sites
Primary reprocessing-waste or high-level-waste sites	4 sites

^a The detailed DOE inventory total depends on the treatment of naval fuel and commercial-origin fuel transferred to DOE custody. The value shown here follows the inventory basis used in this work. Quantities are rounded and may therefore differ slightly among inventory summaries.

The commercial inventory has also changed in composition over the history of the U.S. nuclear power program. In the early decades of light-water-reactor operation, discharged fuel commonly had burnups below 20 GWd/MTU, and initial enrichments were often near 2 wt% ²³⁵U. Improvements in fuel design, reactor operation, and fuel-management strategies allowed utilities to extend cycle lengths and extract more energy from each assembly. By the twenty-first century, average discharge burnup had increased to approximately 50 GWd/MTU, while typical initial enrichment approached 4.5 wt% ²³⁵U [1].

Figure ?? shows the increase in annual average burnup and initial enrichment for U.S. light-water-reactor discharges between 1968 and 2022. Although pressurized-water-reactor and boiling-water-reactor fuel follow somewhat different histories, both exhibit the same broad trend toward higher enrichment and higher discharge burnup.

Higher burnup changes the isotopic composition of discharged fuel. During irradiation, neutron capture in ²³⁸U produces plutonium isotopes, while successive neutron-capture and decay reactions generate neptunium, americium, and curium. Extending irradiation therefore generally increases the production of plutonium and minor actinides within an assembly, although the exact inventory depends on enrichment, reactor type, neutron spectrum, power history, and cooling time. The modern UNF inventory is consequently not uniform:

assemblies discharged in different decades contain different quantities of fissile material, transuranic elements, fission products, and decay heat[1].

These compositional differences are important because the nuclides that control UNF behavior change with cooling time. Shortly after discharge, fission products dominate decay heat. In particular, ^{137}Cs and ^{90}Sr , together with their decay products, are major contributors over the first several decades. Their heat output decreases substantially as they decay. Over longer periods, transuranic isotopes—especially plutonium and americium isotopes—become the dominant contributors to the remaining heat and radiotoxicity [1].

This transition is significant for waste-management strategy. Storage can allow short-lived fission products to decay, but storage alone does not alter the fundamental long-term burden posed by plutonium and the minor actinides. These nuclides must otherwise remain isolated while they decay over much longer timescales. In contrast, actinides can be separated and irradiated in a suitable neutron spectrum, where fission converts them into shorter-lived fission products while releasing additional energy. Long-lived fission products may also be candidates for specialized transmutation, but the actinides are the principal component of UNF that can be destroyed through fission rather than managed solely by long-term storage and geologic isolation[9].

The increasing mass, geographic dispersion, and transuranic content of the U.S. UNF inventory therefore motivate the fuel-cycle simulations in this work. A realistic assessment of accelerator-driven transmutation must represent not only the total mass of available fuel, but also the differences among assemblies in discharge date, burnup, enrichment, location, and isotopic composition. These attributes determine which portions of the inventory can be selected, separated, and supplied to an accelerator-driven system.

2.2 The Cyclus Fuel Cycle Simulation Framework

2.2.1 Framework Basics

CYCLUS is an open-source, agent-based nuclear fuel cycle simulation framework designed to represent the deployment, operation, and interaction of nuclear fuel cycle facilities over time. The framework tracks discrete facilities and material resources rather than representing the fuel cycle solely through aggregated fleet-level flows. This design supports analyses in which individual facility behavior, material availability, and institutional decisions influence system-level outcomes [10].

Agents in a CYCLUS simulation are organized into three hierarchical types: regions, institutions, and facilities. Regions form the highest organizational level and may represent countries, geographic areas, or other broad groupings. Institutions operate within regions and manage the deployment and retirement of facilities. Facilities represent physical or functional components of the nuclear fuel cycle, including mines, enrichment plants, reactors, separations facilities, storage facilities, and repositories [10].

A CYCLUS simulation advances through discrete time steps, commonly defined as months. During each time step, facilities may request resources, offer available resources, conduct material transactions, process their inventories, and update their internal states. Modeling facilities and time discretely allows the framework to capture effects such as asynchronous reactor refueling, temporary shortages of fuel-cycle materials, and differences in facility deployment and retirement schedules[11].

The behavior of each agent is defined by an *archetype*. Archetypes are modular software components that implement the physical behavior, operating logic, or decision rules of a region, institution, or facility. They are compiled independently from the CYCLUS kernel and loaded dynamically at the beginning of a simulation. This plug-in architecture allows alternative models of the same fuel-cycle process to be exchanged without

modifying the core simulation framework [10], [12].

The CYCLUS kernel provides shared simulation services, including time management, agent deployment, resource tracking, data recording, and resource exchange. Archetypes access these services through the agent application programming interface. As illustrated in Figure ??, the kernel provides the central simulation environment, the agent interface connects the kernel to agent models, and independently developed archetype libraries provide the behavior of the simulated regions, institutions, and facilities.

2.2.2 Materials and Decay

Resources exchanged in CYCLUS are represented as discrete objects. The two primary resource types are *materials* and *products*. A material resource has an associated quantity and isotopic composition, whereas a product represents a non-nuclear resource whose quality may be described without a nuclide vector. Nuclear fuel cycle simulations therefore generally use material resources to represent fresh fuel, spent nuclear fuel, separated actinides, and radioactive waste [10].

A CYCLUS material object contains a total mass and a *Composition* object. The composition stores the relative abundance of the nuclides present in the material and is treated independently from the material quantity. This separation allows multiple material objects to reference the same composition and permits operations such as splitting, combining, and transmuting material while preserving mass accounting. Materials may also be stored in facility inventories and exchanged between agents through the resource exchange system.

Radioactive decay can be applied directly to material resources within CYCLUS. When a material is decayed over a specified time interval, its nuclide vector is updated according to the radioactive decay chains available to the framework. The decay calculation uses nuclear data supplied through PyNE, including half-lives, decay constants, branching relationships, and daughter products. The total quantity of a material remains associated with the resource while its isotopic composition changes with time [10].

Decay in the simulations developed for this work is handled at the simulation level through the CYCLUS control block, with decay set to `lazy`. Under this setting the kernel decays a material’s composition on demand rather than at every time step, computing the decayed nuclide vector when a composition is next inspected and caching the result for reuse [10]. This choice retains the physical effect of radioactive decay over the multi-decade simulation horizon while avoiding the cost of decaying every inventory at every step. Decay is significant over this horizon: the roughly 14-year half-life of ^{241}Pu , for example, produces appreciable in-growth of ^{241}Am across a thirty-year run, altering the minor actinide vector that the inventory and accelerator-driven system models are designed to track.

2.2.3 The Dynamic Resource Exchange

Transactions among CYCLUS facilities are coordinated through the dynamic resource exchange (DRE). The DRE provides a general mechanism through which independently developed archetypes can exchange resources without requiring direct knowledge of one another’s internal implementation. Facilities communicate their demand and supply to the framework, and the exchange system determines a feasible set of transactions for each time step [10].

The exchange begins when consuming facilities generate requests for resources. A request identifies a desired commodity, quantity, and, when relevant, a target material composition or other resource quality. Supplying facilities evaluate these requests and generate bids describing the resources they are able to provide.

A bid links a particular offered resource to a request and may provide all or only part of the requested quantity[11].

Facilities may also assign preferences to possible trading partners or resources. Preferences allow a requester to rank otherwise feasible bids, for example by favoring a particular supplier, fuel composition, or material characteristic. In addition, suppliers and requesters may impose constraints such as production capacity, inventory availability, throughput, or minimum and maximum transaction quantities. These constraints define which combinations of trades are physically or operationally feasible.

The requests, bids, preferences, and constraints form a resource assignment problem. CYCLUS formulates this assignment as an optimization problem, solvable in either a linear or a mixed-integer form, and the DRE solves it to select a set of transactions that satisfies the applicable constraints while favoring higher-preference trades [11]. After the solution is determined, the participating facilities execute the selected transactions and update their inventories. This process is repeated at each simulation time step, allowing supply and demand relationships to change dynamically as facilities are deployed, retired, depleted, or otherwise alter their behavior.

By separating exchange coordination from facility-specific behavior, the DRE allows archetypes to be developed and combined modularly. A storage facility, reactor, separations plant, or repository can participate in the same exchange system as long as it implements the appropriate request, bidding, and trade interfaces. This separation is particularly important for this work because the inventory and accelerator-driven system models must interact with both existing and newly developed fuel cycle facilities.

2.2.4 Extensions Developed in This Work

This work extends the CYCLUS ecosystem and its `cycamore` archetype library [13] with two facility archetypes developed within the `einstein` archetype library. The `us_inventory` facility represents the U.S. spent nuclear fuel inventory at assembly resolution, reading isotopic compositions and assembly metadata from the STANDARDS 5.0.1 dataset[4], [5] and supplying material according to user-selected inventory selection policies. The `ads` facility represents an accelerator-driven subcritical burner, transforming a loaded core composition using a library of pre-computed Serpent depletion cases. Both archetypes participate in the resource exchange described above: `us_inventory` as a supplier whose selection policies determine which assemblies it offers against a given request, and `ads` as a requester whose feed preferences rank inventory and separations output. Their design, implementation, and the four simulation scenarios in which they are exercised are described in Chapter 3.

2.3 Accelerator-Driven Systems

ADSs combine a high-power particle accelerator with a subcritical nuclear assembly. They have been proposed as dedicated systems for the transmutation of plutonium, minor actinides, and selected long-lived fission products recovered from spent nuclear fuel [9]. The ADS concept is particularly relevant to this work because it provides a potential pathway for consuming minor-actinide-rich materials that present significant neutronic and safety challenges in critical reactors.

2.3.1 System Components and Operation

An ADS consists of three principal components: a high-energy proton accelerator, a spallation target, and a subcritical multiplying core. The accelerator produces a proton beam, often with an energy of several hundred megaelectronvolts to approximately one gigaelectronvolt. The beam strikes a heavy-metal target, where high-energy proton–nucleus interactions release neutrons through the spallation process. These source neutrons enter the surrounding subcritical core and induce fission reactions in the fuel [14], [15].

The target must generate a sufficiently intense neutron source while withstanding substantial beam heating, radiation damage, and mechanical loading. Heavy elements are attractive target materials because they produce relatively large numbers of neutrons when bombarded by high-energy protons. Proposed ADS targets include solid tungsten targets and liquid-metal targets based on lead or lead–bismuth eutectic. In some designs, a circulating liquid-metal target also contributes to heat removal [14], [15].

The core is designed to remain subcritical, such that

$$k_{\text{eff}} < 1, \quad (2.1)$$

where k_{eff} is the effective neutron multiplication factor. Unlike a critical reactor, the neutron population in an ADS core cannot sustain itself indefinitely without the external spallation source. Source neutrons initiate fission reactions and are multiplied within the subcritical core, allowing the thermal power produced by fission to exceed the power delivered by the proton beam [15], [16].

The accelerator therefore provides an additional mechanism for controlling the fission power. When the beam is interrupted, the external source of spallation neutrons is removed and the source-driven fission power decreases rapidly. This behavior is sometimes described by stating that the beam acts as an on–off switch for the fission chain. However, turning off the beam does not eliminate radioactive decay heat. The fuel and structures continue to generate heat after shutdown and require reliable residual heat removal [16].

Figure 2.1 illustrates the relationship among the accelerator, spallation target, and subcritical core.

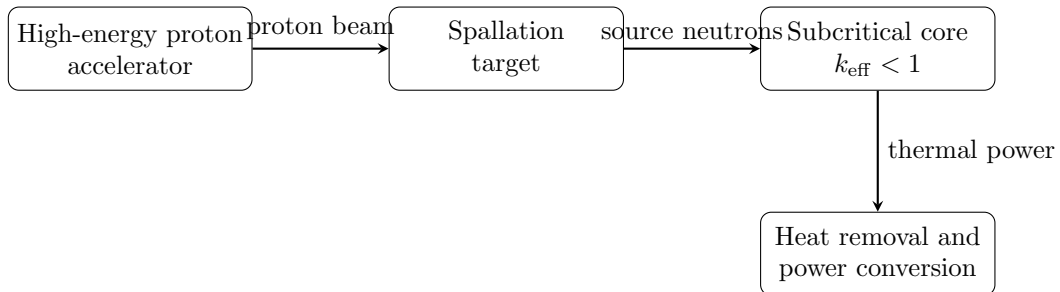


Figure 2.1: Principal components and energy flows of an accelerator-driven subcritical system. This original schematic is based on the ADS descriptions presented in [14], [15].

Accelerator-driven transmutation of used nuclear fuel received sustained attention in the United States during the late 1990s and early 2000s, most prominently through the Accelerator Transmutation of Waste (ATW) program, which developed system scenarios and integration roadmaps for dedicated subcritical burners [17]. Interest in the approach has since re-emerged in the context of managing the growing national inventory.

2.3.2 Minor-Actinide Transmutation

The principal motivation for using ADS technology for nuclear waste management is its ability to accommodate fuels containing large fractions of minor actinides. The minor actinides most commonly considered for transmutation are neptunium, americium, and curium. These nuclides contribute to the long-term radiotoxicity and decay heat of spent nuclear fuel. Their separation and irradiation in a dedicated transmutation system could reduce the quantity of long-lived transuranic material requiring disposal [9].

Fast neutron spectra are generally preferred for minor-actinide transmutation because fission becomes more competitive with radiative capture for many transuranic isotopes at higher neutron energies. A fast-spectrum system can therefore destroy minor actinides through fission rather than merely converting them into heavier actinides [9].

However, loading a critical fast reactor with a high concentration of minor actinides introduces important safety and control challenges. Minor-actinide-rich fuel generally has a smaller effective delayed-neutron fraction than conventional uranium- or plutonium-bearing fuel. Although delayed neutrons represent only a small fraction of the total fission neutron population, they substantially slow the response of a critical reactor to reactivity changes. A reduction in the effective delayed-neutron fraction therefore causes the reactor power to respond more rapidly to disturbances [16].

Minor-actinide loading may also reduce the magnitude of negative Doppler reactivity feedback. In conventional reactor fuel, increasing fuel temperature broadens neutron-absorption resonances and generally introduces negative reactivity. This prompt feedback opposes the temperature increase. The magnitude of this effect can be weaker in minor-actinide-rich, fast-spectrum fuels because of their isotopic composition and the reduced importance of resonance absorption in a fast spectrum [9], [16].

The combination of a reduced delayed-neutron fraction and weaker negative Doppler feedback makes a critical core with a high minor-actinide content more difficult to control. Critical fast-reactor concepts may therefore limit the amount of minor actinides loaded in the core or distribute them among fuel containing larger quantities of uranium or plutonium [16].

Subcritical operation provides an alternative approach. Because the core remains below criticality and relies on an external neutron source, an ADS can accommodate fuel compositions that would be difficult to use at high concentrations in a critical reactor. Subcriticality provides additional margin against prompt criticality and separates, to some extent, the control of the neutron source from the inherent reactivity of the fuel [16].

Subcriticality does not eliminate the need to evaluate reactivity accidents, cooling failures, material behavior, or decay heat removal. Nevertheless, it relaxes an important neutronic constraint on the amount of minor actinides that can be loaded into a dedicated transmutation system. This capability is the central reason that ADS technology is relevant to this thesis. The purpose of the modeled fuel cycle is to connect separated material from the U.S. spent nuclear fuel inventory with systems capable of consuming minor-actinide-rich streams that may not be suitable for concentrated use in critical reactors.

2.3.3 Target and Coolant Options

ADS concepts differ in their target materials, coolants, fuel forms, neutron spectra, and power levels. Lead and lead-bismuth eutectic are among the most frequently proposed heavy liquid metals because their large atomic masses support spallation neutron production and their high boiling temperatures permit low-pressure primary systems [14], [15].

Lead–bismuth eutectic has a substantially lower melting temperature than pure lead, reducing some of the operational challenges associated with coolant freezing. The most advanced current example is Multi-purpose Hybrid Research Reactor for High-tech Applications (MYRRHA), a lead–bismuth-cooled subcritical system under development at the Belgian Nuclear Research Centre. Its first phase, the MYRRHA Injector and low-energy accelerator (MINERVA) proton-accelerator and target facility, began construction in 2024 [18], making MYRRHA the closest existing demonstration of the accelerator-driven concept. However, neutron activation of bismuth produces polonium-210, which creates radiological and maintenance concerns. Both lead and lead–bismuth can also cause corrosion or erosion of structural materials. Their use therefore requires careful control of coolant chemistry, oxygen concentration, temperature, and flow conditions [15].

Pure lead avoids the production of polonium associated with bismuth but has a higher melting temperature. This increases the importance of freeze prevention during startup, shutdown, and off-normal operation. Despite this challenge, lead has been proposed both as a coolant and as a carrier medium for mobile-fuel ADS concepts [15], [19].

The ADS concept that provides the initial technical basis for the EINSTEIN project was developed to transmute material derived from the U.S. spent nuclear fuel inventory. The proposed system uses a high-energy proton accelerator, a spallation neutron source, and a lead-based subcritical system. The fuel concept contains plutonium and minor actinides in a mobile lead-based medium, permitting continuous or periodic processing of the fuel inventory [19].

Gas-cooled ADS concepts have also been investigated. Helium is chemically inert and avoids many of the corrosion and activation issues associated with heavy liquid metals. However, its low density and volumetric heat capacity require high-pressure operation and significant pumping power. The coolant and target selection therefore influences neutron production, material compatibility, plant efficiency, operating pressure, safety behavior, and the design of the primary heat-removal system [15].

2.3.4 Technical and Economic Limitations

Despite their potential advantages for transmutation, ADS facilities face substantial technical and economic challenges. Accelerator availability is one of the most important engineering concerns. A sudden beam interruption causes a rapid decrease in fission power and temperature. Restoration of the beam causes the power and temperature to increase again. Repeated beam interruptions can therefore impose cyclic thermal loads on the fuel, spallation target, coolant systems, and structural components [20].

The Organisation for Economic Co-operation and Development (OECD) Nuclear Energy Agency beam-interruption benchmark specifically examined the temperature transients caused by beam trips in a lead–bismuth-cooled, subcritical system. The study identified thermal fatigue of structural components as an important consequence of repeated beam interruptions [20]. More recent analysis of the China Initiative Accelerator Driven System has also evaluated fuel cladding fatigue under repeated beam-trip transients [21].

Producing a continuous, high-current proton beam with the availability required for a power-producing nuclear system is more demanding than operating an accelerator intended for intermittent research use. Fault-tolerant accelerator designs, redundant components, and rapid recovery procedures may reduce the number and duration of beam interruptions, but these features add complexity and cost [15].

The spallation target presents additional engineering challenges. It must tolerate concentrated beam heating, radiation damage, gas production, thermal stresses, and interactions with the coolant. Target replacement, inspection, and maintenance must also be performed in a highly radioactive environment [15].

ADS deployment additionally requires supporting fuel-cycle infrastructure. Spent nuclear fuel must first

be processed to recover the actinides intended for transmutation. The resulting minor-actinide-rich material must then be fabricated into fuel or introduced into an appropriate mobile-fuel medium. After irradiation, the material may require repeated separation and recycle to achieve substantial actinide destruction. These activities introduce material losses, secondary waste streams, safeguards requirements, worker exposure, and additional facilities[9].

An ADS also combines the capital cost and regulatory requirements of a nuclear reactor facility with those of a high-power accelerator. Furthermore, the accelerator consumes a portion of the electricity generated by the system. Comparative OECD/Nuclear Energy Agency (NEA) studies have therefore emphasized that the potential transmutation benefits of ADS technology must be evaluated alongside its fuel-cycle requirements, technical complexity, research and development needs, and economic performance[16].

These limitations make a system-level fuel-cycle analysis necessary. The performance of an ADS cannot be judged solely from the rate of transmutation inside its core. A complete assessment must also consider the availability and composition of feed material, separations capacity, facility deployment, accelerator and reactor throughput, recycle requirements, and the production of secondary waste streams. These interactions motivate the CYCLUS simulation models developed in this work.

2.4 The EINSTEIN Program

The EINSTEIN project is part of the ARPA-E NEWTON program. The NEWTON program was established to support technologies that enable the transmutation of used nuclear fuel and reduce the burden placed on permanent waste-disposal facilities [3]. The EINSTEIN project is led by the University of Illinois Urbana-Champaign [22].

NEWTON addresses the transmutation problem through three complementary technical categories. Category A focuses on technologies for generating and accelerating particle beams capable of initiating transmutation reactions. Category B focuses on the design, fabrication, and processing of targets or fuel forms containing the nuclides to be transmuted. Category C integrates the technologies developed in Categories A and B through system-level modeling, life-cycle analysis, techno-economic analysis, and databases of materials and components [3], [23]. UIUC is a Category C and therefore does not develop a single accelerator or transmutation target in isolation. Instead, it provides a common framework for comparing candidate systems and identifying the technical and economic conditions under which they could contribute to management of the U.S. used nuclear fuel inventory.

Within the EINSTEIN project, the term XDS refers collectively to subcritical transmutation systems sustained by an external neutron source. Within this framing, ADSs use an accelerator and spallation target to produce the source neutrons, while fusion-driven systems (FDSs) use neutrons generated by a fusion source. The two classes may differ substantially in neutron energy, source geometry, target and coolant materials, reliability constraints, and technology readiness. Nevertheless, both couple an external neutron source to a subcritical multiplying region and can therefore be represented within a common fuel-cycle and assessment framework. The initial EINSTEIN analyses include a high-energy, proton-driven, lead-cooled ADS and a 14.1 MeV fusion-neutron-driven, salt-cooled system [24].

The central objective of EINSTEIN is to develop a data-informed framework for rapidly assessing accelerator- or fusion-driven system (XDS) technologies according to both their transmutation performance and their financial implications. The project connects nuclear data, neutronics, materials compatibility, fuel-cycle behavior, and economic performance rather than evaluating each component independently. The

official project description identifies the development of a comprehensive life-cycle analysis framework spanning the full transmutation process, from the composition of the incoming used-fuel stream through irradiation, processing, and the resulting waste or product streams [22]. Experimental and computational activities are used to reduce uncertainties in neutron transport, material properties, and component performance.

The UIUC project team is organized around several interacting technical areas. Nuclear-data and diagnostics work identifies reactions whose uncertainties significantly affect neutron multiplication and transmutation predictions and develops experimental strategies to reduce those uncertainties. Neutronics analysis provides source-term, criticality, depletion, and transmutation calculations for selected XDS concepts. Materials research evaluates corrosion, irradiation damage, thermophysical properties, and the chemical behavior of lead coolants and molten salts. Experimental work investigates the stability and solubility of representative actinide surrogates and the compatibility of structural materials with the candidate coolant and fuel media [24].

These activities supply the inputs required for the fuel-cycle and techno-economic assessment. The fuel-cycle portion of the project represents the U.S. used nuclear fuel inventory, separations facilities, XDS facilities, recycle pathways, storage facilities, and waste streams in CYCLUS. The inventory characterization begins with national spent-fuel records and age-corrected isotopic compositions, which are used to define realistic feed streams for the modeled XDS technologies. New CYCLUS archetypes are then used to evaluate how the choice of feed material, facility throughput, deployment schedule, separations capacity, and material-selection policy affects system-level transmutation performance [24].

The techno-economic analysis uses the outputs of the fuel-cycle and component models to examine capital costs, operating costs, potential revenues, return on investment, and the sensitivity of economic conclusions to uncertain design parameters. This coupling is important because a design that performs well in a stand-alone neutronics calculation may still require an impractical number of facilities, excessive separations capacity, costly transportation, or unavailable supporting infrastructure. Conversely, a concept with a lower per-facility transmutation rate may be more attractive if it is easier to deploy, operate, and integrate with the existing nuclear fuel cycle.

The work presented in this thesis lies primarily within the fuel-cycle modeling portion of EINSTEIN. It develops the `us_inventory` and `ads` archetypes in the `einstein` library and uses them to connect the characteristics of the national used-fuel inventory with candidate XDS deployment scenarios. The resulting simulations provide material-flow, facility-scaling, and inventory-depletion information that can be combined with the neutronics, materials, and economic analyses performed elsewhere in the project. In this way, the thesis contributes the system-level link between the physical design of an XDS and its potential effect on the U.S. used nuclear fuel inventory.

2.5 Prior Fuel Cycle Simulations of Transmutation Systems

Research on accelerator-driven transmutation has progressed through several distinct stages. Early studies established the physics of coupling an intense accelerator-produced neutron source to a subcritical multiplying medium. Subsequent programs developed and tested individual enabling technologies, including high-current accelerators, liquid-metal spallation targets, subcriticality-monitoring methods, minor-actinide fuels, and lead or lead-bismuth coolant systems. More recent studies have examined how dedicated transmutation facilities could be incorporated into regional or national nuclear fuel cycles.

These bodies of work do not all address the same modeling question. Reactor physics studies generally

evaluate neutron multiplication, power distributions, safety coefficients, and nuclide destruction within one core. Technology demonstrators test particular components or operating procedures. Fuel-cycle scenario studies instead examine the deployment of many reactors and supporting facilities over several decades, tracking quantities such as plutonium and minor-actinide inventories, separations requirements, fuel fabrication capacity, and waste production. The present work belongs primarily to the third category, but its facility models are informed by the first two.

2.5.1 European Demonstrator Programs

Europe has sustained the most continuous multinational program of ADS research. Its development pathway has included conceptual system studies, critical and subcritical experiments, a megawatt-class liquid-metal spallation-target experiment, and the staged development of the MYRRHA research facility. These programs have demonstrated important elements of an ADS, although no European facility has yet operated as a high-power, minor-actinide-burning ADS.

One influential early European concept was Carlo Rubbia’s Energy Amplifier. The Energy Amplifier combined a high-energy proton accelerator, a heavy-metal spallation target, and a subcritical fast-spectrum blanket. Early versions emphasized energy production from thorium, while later analyses also considered waste transmutation and fast-spectrum operation [25]. The importance of this work was not that it produced a directly deployable fuel-cycle design, but that it articulated a complete accelerator–target–blanket system and stimulated coordinated European research on spallation targets, lead-based coolants, subcritical reactor physics, and partitioning and transmutation.

European research was subsequently coordinated through programs including EUROTRANS, which investigated an experimental ADS and the European Facility for Industrial Transmutation European Facility for Industrial Transmutation (EFIT). EFIT was conceived as a lead-cooled, fast-spectrum minor-actinide burner rather than as a general-purpose power reactor. Its design studies addressed accelerator requirements, target integration, subcritical-core design, dedicated minor-actinide fuel, and the supporting closed fuel cycle. In the “double-strata” strategy commonly associated with these studies, conventional critical reactors would manage most uranium and plutonium recycling, while a smaller fleet of dedicated ADS facilities would receive and destroy concentrated minor-actinides[26].

The EFIT studies are particularly relevant to fuel-cycle analysis because they treated an ADS not as an isolated core but as one component of a broader fleet. Regional scenario calculations examined the number of ADS facilities required to stabilize or reduce minor-actinide inventories and estimated the associated separations and fuel-fabrication capacities. These analyses demonstrated that transmutation performance depends not only on the destruction rate of one ADS, but also on cooling delays, reprocessing losses, fuel-fabrication throughput, deployment dates, and the availability of separated feed material [26].

The principal European engineering demonstrator is MYRRHA developed by Belgian Nuclear Research Centre (SCK CEN) in Belgium. MYRRHA is designed as a lead–bismuth-eutectic-cooled, fast-spectrum research system capable of subcritical accelerator-driven operation and, in the broader design concept, critical operation. Its purposes include demonstrating ADS technology, conducting fuels and materials irradiations, supporting radioisotope production, and providing experimental data relevant to minor-actinide transmutation [14], [18].

MYRRHA is being implemented in phases. The first phase, MINERVA, comprises the lower-energy portion of the proton accelerator and associated target stations. Construction of this phase began in 2024. The later phases would extend the accelerator and couple it to the subcritical reactor. MYRRHA therefore

represents the most advanced European program directed toward a power-scalable ADS, but it should not yet be described as an operating accelerator-driven reactor [18].

Several predecessor experiments were developed to reduce the technical risk of MYRRHA. The Generator of Uninterrupted Intense Neutrons at the Lead VENUS Reactor (GUINEVERE) project coupled the Générateur de Neutrons Pulsés Intenses, third generation with continuous beam (GENEPI-3C) deuteron accelerator to the fast-spectrum configuration of the Vulcan Experimental Nuclear Study reactor (VENUS-F) fast lead critical and subcritical assembly at SCK CEN. GUINEVERE was designed principally to investigate reactor-physics questions, including the determination and online monitoring of subcriticality, source-jerk and beam-interruption methods, core loading procedures, and the transition between critical and subcritical configurations [27], [28], [29]. It was a low-power experimental facility rather than a fuel-burning demonstrator, but it provided evidence that an accelerator could be coupled to a fast subcritical assembly and that the system’s reactivity could be measured using realistic source-operation modes.

A separate enabling-technology experiment was the MEGAwatt Pilot Experiment, or MEGAwatt Pilot Experiment (MEGAPIE), operated at the Swiss spallation neutron source Swiss Spallation Neutron Source (SINQ) in 2006. MEGAPIE exposed a circulating lead–bismuth target to a continuous 590 MeV proton beam at approximately 0.77 MW. It was the first liquid-metal spallation target operated in the megawatt beam regime and supplied experimental data on neutron production, heat removal, liquid-metal operation, activation products, structural behavior, and post-irradiation examination [30], [31], [32]. MEGAPIE did not include a multiplying reactor core, so it did not demonstrate an integrated ADS. Its importance was instead the validation of one of the most demanding ADS components under an irradiation environment approaching that required for a full-scale system.

France contributed to these European programs through Commissariat à l’Énergie Atomique et aux Énergies Alternatives (CEA), National Centre for Scientific Research (CNRS), and other institutions, as well as through national fuel-cycle scenario studies. CEA developed the Commellini-Sicard (COSI) simulation code to model reactor fleets and associated fuel-cycle facilities over long time horizons. COSI studies examined transitions from light-water reactors to sodium-cooled fast reactors and compared homogeneous, heterogeneous, and dedicated minor-actinide transmutation strategies [33]. Other French and European studies compared ADS-based transmutation with minor-actinide burning in critical fast reactors, including differences in fuel-cycle cost, inventory reduction, and supporting facility requirements [34], [35].

Taken together, the European programs established a progression from component experiments to integrated demonstrator design and fleet-level scenario analysis. GUINEVERE addressed source coupling and subcriticality monitoring, MEGAPIE addressed a high-power liquid-metal target, EFIT provided a reference industrial transmuter design, and MYRRHA seeks to integrate many of these capabilities into a research facility. Their scenario studies also showed that the viability of ADS deployment cannot be inferred from single-core transmutation efficiency alone.

2.5.2 United States Transmutation Programs, 1990s–2000s

Modern U.S. accelerator-driven transmutation work grew partly from studies at Los Alamos National Laboratory in the late 1980s and early 1990s. Bowman et al. proposed coupling an intense accelerator-produced neutron source to a subcritical multiplying and transmuting medium. The concept considered the destruction of transuranic elements and selected long-lived fission products, together with energy production from fertile material [36]. This work helped establish the technical foundation for the (ATW) program.

ATW was developed during the 1990s as a system for treating commercial spent nuclear fuel using

partitioning, accelerator-driven transmutation, and repeated recycle. A reference ATW complex included front-end separations, minor-actinide and plutonium fuel preparation, one or more accelerator-driven subcritical burners, recycle separations, and waste conditioning. The U.S. Department of Energy submitted a development roadmap to Congress in 1999, identifying required research in accelerators, spallation targets, separations, fuels, materials, reactor physics, safety, and systems analysis [17], [37].

The roadmap treated ATW explicitly as a fuel-cycle system rather than merely as a reactor. This was important because repeated recycle is required to achieve high actinide destruction. No irradiation cycle destroys every atom fed to the system: some actinides remain in the discharged fuel, some are converted into other actinides, and small fractions are lost during chemical processing and fabrication. Consequently, the overall reduction in waste burden depends on the efficiency and capacity of the complete recycle loop.

The follow-on Advanced Accelerator Applications (AAA) program broadened the scope of the U.S. effort to include waste transmutation, energy applications, and enabling accelerator technologies. Program planning documents considered the reliability of high-power proton accelerators, liquid-metal and solid targets, subcritical blanket configurations, separations based on aqueous and pyrochemical methods, and fabrication of fuels containing high fractions of transuranic elements. Much of this work remained at the level of design studies, laboratory research, and component development. No integrated U.S. ATW or AAA engineering demonstrator was constructed.

Systems studies performed for ATW and related programs compared combinations of light-water reactors, critical fast reactors, and accelerator-driven systems. These studies investigated questions such as whether plutonium should first be recycled in light-water reactors, whether ADS facilities should consume all transuranics or only minor actinides, and how many recycle passes would be required. Their results repeatedly showed that conclusions were sensitive to separations efficiency, fuel cooling time, accelerator availability, support ratio, and the assumed growth or decline of nuclear power [17], [37].

After the large ATW and AAA programs ended, U.S. ADS work continued through more focused reactor-physics, materials, target, and systems studies. A recent and particularly relevant example is the Argonne National Laboratory concept reported by Gohar, Cao, and Kraus. Their design was developed specifically to consume plutonium and minor actinides derived from the U.S. spent nuclear fuel inventory. It employs a 1 GeV proton beam, a lead-based spallation target, and a subcritical fast blanket with mobile fuel containing plutonium and minor actinides [19].

Gohar et al. used Monte Carlo transport and depletion calculations to examine several blanket compositions and estimate the actinide consumption achievable over decades of full-power operation. The study concluded that a small fleet of large ADS units could consume the minor-actinide inventory assumed in its analysis. However, its scope was a set of reference ADS designs and their depletion performance. It did not dynamically model the assembly-level U.S. inventory, site-to-site material movement, separations deployment, facility-construction schedules, or competition among facilities for particular spent-fuel resources. Thus, it provides the principal reactor design basis for the ADS model in this thesis, but not the system-level model needed to determine how that design would be supplied and deployed.

2.5.3 Ongoing Construction: CiADS

China began a national ADS development program through the Chinese Academy of Sciences in 2011. Early work included the (China Lead-based Reactor (CLEAR)) program and a series of lead-bismuth experimental loops, zero-power facilities, and non-nuclear test systems. These activities supported the development of coolant technology, structural materials, fuel assemblies, components, and operating methods for a lead-cooled

subcritical system [38].

The China initiative Accelerator Driven System (CiADS), led by the Institute of Modern Physics of the Chinese Academy of Sciences, is now under construction in Huizhou. The integrated facility is designed to couple a superconducting proton linear accelerator, a lead–bismuth spallation target, and a lead–bismuth-cooled subcritical reactor. Published design studies describe a first-stage reactor with a maximum thermal power of approximately 10 MW. The initial core uses oxide fuel and is intended primarily to demonstrate the engineering feasibility, control, safety, and coupled operation of the ADS rather than to serve immediately as an industrial-scale minor-actinide burner [39], [40].

CiADS is significant because it moves beyond isolated component tests toward construction of an integrated accelerator–target–subcritical-reactor facility. Its research program addresses accelerator reliability, beam transport, target behavior, lead–bismuth technology, reactor control, thermal hydraulics, and protection systems. In the proposed operating strategy, core power is controlled through the proton-beam power because the subcritical core does not contain a conventional critical-reactor control system for routine power regulation [39], [41].

Dynamic and transient studies of CiADS have analyzed beam trips, beam overpower, loss of flow, loss of heat sink, and station-blackout conditions. These calculations indicate substantial thermal margins for the low-power demonstrator while identifying longer-term concerns such as lead–bismuth corrosion, creep, and fatigue under repeated thermal cycling [40]. The project therefore provides an important test of whether the theoretically attractive subcritical characteristics of ADSs can be realized in a maintainable engineering system.

CiADS should not be interpreted as a completed demonstration of a commercial transmutation fuel cycle. The first-stage system is much smaller than the gigawatt-thermal designs commonly assumed in industrial fuel-cycle studies, and its initial mission emphasizes technology demonstration. Nevertheless, operational experience from CiADS could reduce major uncertainties presently treated parametrically in systems models, including beam availability, component lifetime, maintenance frequency, achievable capacity factor, and the coupling between accelerator and reactor operation.

2.5.4 Fuel-Cycle Simulation Studies

The studies described above span a wide range of spatial and temporal resolutions. Core-design calculations may represent individual fuel pins and detailed nuclide vectors but usually consider only one reactor and a limited number of irradiation cycles. Fuel-cycle simulators generally model decades of fleet evolution, but they aggregate fuel into batches, reactor types, or material streams. This trade-off between reactor detail and system breadth is central to understanding the gap addressed by this thesis.

The OECD/NEA regional-scenarios study extended this type of analysis to groups of countries with different nuclear policies and fuel-cycle infrastructure. It used the EFIT design as the reference ADS and calculated the number of transmuters and associated fuel-cycle facilities required to stabilize minor-actinide inventories across a regional system [26]. This work showed that regional sharing of specialized facilities could reduce duplication of expensive separations and fabrication plants. However, its material representation was at the level of national or regional inventory streams rather than individual discharged assemblies.

France’s COSI code has been used extensively for dynamic scenario analysis. COSI represents reactor fleets, fuel fabrication, irradiation, cooling, reprocessing, storage, and waste production over time. Applications have included transitions from the existing French light-water-reactor fleet to sodium-cooled fast reactors, along with comparisons of minor-actinide management options [33]. These studies are highly detailed in their

treatment of fleet evolution and industrial fuel-cycle capacity, but they generally use reactor and fuel batches characterized by representative compositions rather than a geospatially explicit assembly inventory.

European projects also compared the economic and material consequences of minor-actinide transmutation in critical fast reactors and ADS facilities. Chabert et al. compared no-transmutation, fast-reactor, and ADS options, while Merino-Rodríguez et al. cross-checked economic and mass-balance calculations for scenarios involving dedicated ADS burners [34], [35]. These studies demonstrated that ADS-based strategies may achieve concentrated minor-actinide destruction but incur additional costs associated with the accelerator, dedicated fuel fabrication, and supporting recycle facilities.

The Belgian Analysis of Nuclear-fuel-cycles Including Chain and Cost Assessment (ANICCA) code provides a more recent example directly related to ADS deployment. ANICCA is a dynamic nuclear-fuel-cycle simulator developed to study Belgian fleet evolution and waste management. A published verification and case study examined the effect of adding an ADS to a Belgian nuclear phase-out scenario. The model tracked fleet-level materials and showed the potential reduction of minor-actinide inventories, but it did not model individual commercial fuel assemblies or source-site-specific selection [42].

Other studies have focused on detailed depletion within an ADS rather than deployment of the surrounding fuel cycle. These calculations optimize core geometry, fuel loading, neutron spectrum, or recycle strategy and report nuclide-specific transmutation rates. They are essential for supplying performance parameters to a system model, but they do not themselves answer how quickly suitable material becomes available, where it is stored, which assemblies are selected, or how infrastructure limits deployment [19].

Across the literature reviewed here, system-level ADS studies generally use one of three material representations. The first is an equilibrium mass-flow model, in which annual production and destruction rates are balanced without explicitly simulating each facility. The second is a dynamic fleet model in which reactor types and fuel-cycle plants are deployed over time and exchange aggregated batches. The third is a detailed core or depletion model in which a single ADS is simulated at high isotopic resolution but the external fuel cycle is prescribed rather than dynamically represented.

These approaches have provided estimates of ADS support ratios, required fleet size, equilibrium actinide inventories, separations capacity, fuel-cycle cost, and repository benefits. They have not generally represented the existing U.S. used-fuel inventory as a set of discrete assemblies carrying site, discharge date, burnup, enrichment, mass, and composition metadata. In the literature reviewed for this thesis, no prior ADS fuel-cycle study was identified that dynamically selected individual assemblies from the site-resolved U.S. inventory and supplied them through separations to parameterized ADS facilities.

That distinction is important for the U.S. case. The national inventory is not an equilibrium material stream produced by a uniform reactor fleet. It is a heterogeneous legacy inventory accumulated over several decades and stored at geographically dispersed operating and shutdown sites. Assemblies differ in age, burnup, enrichment, reactor type, remaining heavy-metal mass, isotopic composition, and storage location. A model that begins with a representative average composition cannot evaluate how alternative inventory-selection policies change the material supplied to separations and transmutation facilities.

The present work addresses this gap by combining an assembly-resolved U.S. inventory facility with dynamically deployed ADS and supporting fuel-cycle facilities in CYCLUS. It does not replace detailed neutronics, depletion, materials, or economic models. Instead, it provides the system layer through which those models can be connected to the actual characteristics and availability of U.S. used nuclear fuel. This representation permits analysis of inventory-selection policies, throughput limits, facility scaling, deployment timing, material availability, and the evolution of the remaining national inventory.

Table 2.2: Prior accelerator-driven transmutation programs and studies addressed different levels of technology and fuel-cycle resolution. Information summarized from [18], [19], [26], [27], [31], [40], [42].

Program or study	Type	Principal scope	Material resolution
GUINEVERE	Experimental facility	Accelerator coupling, fast subcritical physics, and online reactivity monitoring	Core configurations; no fuel-cycle deployment
MEGAPIE	Component experiment	Megawatt-class liquid lead–bismuth spallation target	Target and neutron-source behavior only
MYRRHA	Staged demonstrator	Accelerator, target, fast-spectrum research reactor, fuels, materials, and ADS operation	Core and irradiation experiments; supporting scenario studies
EFIT / European scenarios	Reference design and fleet studies	Dedicated minor-actinide burning and regional deployment	Aggregated fuel batches and regional inventories
ATW / AAA	U.S. program and system studies	Separations, recycle, accelerator, target, and subcritical burner	Process streams and representative fuels
Gohar et al.	ADS design and depletion study	Lead-based mobile-fuel ADS for the U.S. actinide inventory	Prescribed blanket compositions
CiADS	Integrated demonstrator under construction	Coupled accelerator, spallation target, and low-power subcritical reactor	Detailed core and component models
COSI / AN-ICCA studies	Dynamic fuel-cycle simulation	Fleet transitions, facility capacities, inventories, and waste	Reactor fleets and aggregated fuel batches
This work	Dynamic fuel-cycle simulation	U.S. inventory selection, separations, ADS deployment, throughput, and inventory evolution	Individual assemblies with associated meta-data

2.6 Research Gap

The literature reviewed in Section 2.5 demonstrates substantial progress in accelerator-driven transmutation, but the principal areas of progress have remained separated. Reactor-physics and engineering studies have developed increasingly detailed representations of accelerators, spallation targets, subcritical cores, fuels, coolants, and transmutation performance. Fuel-cycle studies have examined the deployment of reactors, separations plants, fabrication facilities, and waste-management infrastructure over long periods. National spent-fuel databases have also developed sufficiently detailed records to characterize individual discharged fuel assemblies. However, these capabilities have not generally been combined in a single simulation. In the literature reviewed for this thesis, no published study was identified that dynamically couples a real, assembly-resolved U.S. spent nuclear fuel inventory to an ADS facility inside a fuel-cycle simulator and evaluates how the selection of particular assemblies affects transmutation-system deployment and the evolution of the remaining national inventory.

European ADS research has produced the most mature combination of experimental facilities, component demonstrations, core calculations, and regional fuel-cycle studies. GUINEVERE investigated accelerator coupling and subcriticality monitoring, MEGAPIE demonstrated a liquid-metal spallation target under a megawatt-class proton beam, and MYRRHA has advanced the engineering design and staged implementation of an integrated ADS research facility [14], [18], [27], [28], [31]. European fuel-cycle tools have also examined fleet transitions and minor-actinide transmutation. For example, COSI has been used to evaluate French reactor and fuel-cycle transitions, while ANICCA has modeled the introduction of an ADS into a Belgian nuclear phase-out scenario [33], [42]. These studies provide detailed information about European reactor fleets, fuel-cycle infrastructure, and transmutation strategies, but their feed materials are represented as reactor-specific or scenario-specific fuel streams and batches. They do not begin with the discrete assembly records of the existing U.S. inventory or evaluate alternative rules for selecting particular U.S. assemblies for processing.

The U.S. Accelerator Transmutation of Waste and Advanced Accelerator Applications programs addressed many policy-relevant aspects of a complete transmutation system. Their roadmaps and system studies considered separations, repeated actinide recycle, fuel fabrication, accelerator reliability, spallation-target design, subcritical burners, and final waste management [17], [37]. These efforts recognized that the performance of an ADS could not be assessed independently from the facilities that prepare, recycle, and manage its fuel. Nevertheless, the principal ATW and AAA studies were conducted before the development of current agent-based fuel-cycle simulators and before the availability of the present assembly-level STANDARDS database. Consequently, their national inventory was represented through assumed aggregate material streams, reference isotopic compositions, and projected production rates rather than through the individual assemblies that constitute the existing U.S. inventory.

More recent U.S. work has improved the physical basis of the ADS itself. Gohar, Cao, and Kraus developed a lead-based ADS concept specifically for consuming plutonium and minor actinides derived from U.S. spent nuclear fuel and used detailed transport and depletion calculations to estimate actinide consumption over the operating life of the system [19]. This study provides the initial ADS design basis used in the present work, but its treatment of the national inventory begins with the total assumed masses of plutonium and minor actinides. It does not model the discharged assemblies from which those materials would be recovered, the order in which assemblies would be selected, or the time-dependent interaction among inventory, separations, and ADS facilities.

CiADS addresses a different part of the problem. Its published analyses focus on the engineering behavior

of an integrated accelerator, spallation target, and low-power subcritical reactor. Studies have examined proton-beam control, reactor transients, beam trips, thermal margins, coolant behavior, and structural fatigue [21], [38], [39], [40]. This work is essential for demonstrating whether ADS technology can be operated reliably, but CiADS is an engineering demonstrator rather than a simulation of national fuel-cycle deployment. Its analyses therefore do not address how a heterogeneous legacy inventory would be selected, transported, separated, and allocated among multiple future transmutation facilities.

Existing fuel-cycle and inventory studies come closest to connecting these areas. The STANDARDS 5.0.1 Database contains detailed information about the spent nuclear fuel (SNF) inventory in the U.S. with full isotopic composition. Peterson et al. described the planned use of Unified Database (UDB) information in system models such as Next-Generation System Analysis Model (NGSAM) and in fuel-cycle transition analyses using ORION (ORION) [5]. Bae et al. subsequently trained a neural network on UDB depletion data and implemented the resulting model in CYCLUS, enabling efficient prediction of spent-fuel compositions for reactor batches with varying burnup and enrichment [43]. These studies established that realistic U.S. data can improve fuel-cycle simulation. The former described broader integration of the database with system and fuel-cycle tools, while the latter reproduced UDB-derived isotopic behavior without exposing or dynamically transacting the underlying private assembly records. Neither work connected an active assembly inventory to an ADS burner or examined how the choice among available assemblies changes the feed supplied to that burner.

The gap addressed in this thesis is therefore narrower than the absence of either detailed ADS models or detailed inventory data. The missing capability is their integration at the system level. This work represents individual U.S. spent-fuel assemblies as selectable resources within an agent-based fuel-cycle simulation. Assembly records derived from STANDARDS 5.0.1 retain attributes relevant to material selection, including available heavy-metal mass, discharge date, burnup, initial enrichment, and isotopic composition [1]. The `us_inventory` archetype supplies these resources through the CYCLUS Dynamic Resource Exchange rather than first collapsing the national inventory into a single average recipe.

The inventory model is coupled to an ADS archetype whose material transformation is based on precomputed Serpent depletion results. The simulation can therefore evaluate multiple inventory-selection policies, including selection by availability, age, burnup, and enrichment, while maintaining material accounting as assemblies are supplied and their available masses are depleted. The results presented in Chapters 3–4 quantify how these policies change the mass and characteristics of material supplied to the modeled fuel-cycle facilities and how they alter the composition of the inventory remaining after transmutation.

This integration does not by itself constitute a complete recommendation for U.S. waste policy. Questions involving facility siting, transportation approval, safeguards, separations readiness, cost, financing, and repository performance require additional models and institutional assumptions. Rather, the contribution of this thesis is to provide a U.S.-specific material-flow foundation on which those analyses can be built. By connecting an assembly-resolved legacy inventory to an ADS representation in CYCLUS, the work enables transmutation scenarios to be evaluated using the actual heterogeneity and availability of U.S. spent nuclear fuel instead of an assumed national-average feed.

Chapter 3

Methodology

methodology.tex

Chapter 4

Results

results go here

Chapter 5

Conclusions

conclusion.tex Conclusion.

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Appendix A

An appendix

Appendix.